PROJECT PROPOSAL

E-VFU: A MULTI-ATTACK VERIFICATION FRAMEWORK FOR VERTICAL FEDERATED UNLEARNING

Course : CSE-472

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Overview

Our project proposes **Enhanced Vertical Federated Unlearning (E-VFU),** an advancement over the recent Varshney et al. (PETS 2025) framework that addresses critical verification gaps in feature-level unlearning for Vertical Federated Learning (VFL) systems.
While Varshney et al. introduce the first knowledge-distillation based feature unlearning approach (VFU-KD), their work relies exclusively on –

(a) Relatively weak membership inference attacks (MIA) for auditing
(b) Lacks analysis of feature entanglement effects
(c) Provides no formal communication complexity bounds within secure aggregation settings.

We will try to add four novel technical contributions that significantly strengthen both the security and utility guarantees:
1. **Stronger adversarial verification** via gradient-based membership inference, forensic unlearning membership attacks (FUMA), and shadow model attacks.
2. **Feature entanglement detection** using Jacobian-based inter-party influence matrices.

Overview

3. **Communication efficiency analysis** within secure aggregation protocols with formal bounds.
4. **Iterative unlearning robustness evaluation** via sequential feature removal experiments
These additions provide evidence that unlearned features no longer influence model predictions, maintain utility across parties despite feature dependencies, and offer practical deployment guidance within privacy-preserving infrastructure.

Our proposal is a verification framework that rigorously audits Varshney et al.'s existing VFU-KD unlearning method using 4 novel components.

(a) **If Expected result is achieved then** , E-VFU successfully verifies Varshney et al.'s VFU-KD unlearning works robustly across 4 metrics.

(b) **If verification fails,** we pivot to E-VFU reveals critical limitations of existing VFL unlearning.

Background and Problem Statement

Background : Vertical Federated Learning (VFL) is increasingly deployed in healthcare, finance, and cross-institutional analytics where organizations hold different features (columns) of the same entity. Also, Modern data protection mandate right to erasure — not just at raw data level, but also from trained ML models . So, In VFL settings, unlearning is crucial.

The Gap: Verification of existing unlearning methids of VFL
Varshney et al. (PETS 2025) Contribution:

- **Output-logit only MIA is weak :** Recent work (EMNLP 2025) shows unlearning-specific membership inference can succeed at 99% accuracy despite "successful" unlearning.
- **Feature entanglement unaddressed:** If features are correlated, removing one may invalidate other parties' encodings but no analysis provided on this.
- **No communication cost analysis :** Exact overhead within secure aggregation (SMPC/SecAgg) settings is unstudied.
- **Single-shot unlearning only:** No evaluation of repeated/iterative unlearning requests

Our Proposal
Phase-0 : Base Architecture Building on VFU-KD

Our approach extends Varshney et al.'s VFU-KD framework.

- Knowledge Distillation Unlearning:

  hinew = Studenti(xi,Teacher embeddings from active party)

- Active party trains student encoders to mimic teacher behavior on retain set only (removing target set from training). Passive parties freeze encoders; active party retrains fusion layers.

- Assumption: Unlearing epochs fixed at uep=10

Our Proposal
Phase-1: Robust Verification

1. **Gradient-Based Membership Inference Attack**

Motivation: Output logits alone leak limited information; gradients reveal feature structure .

Method: Adapt gradient-based reconstruction attacks (Geiping et al., 2020) to VFL:

1. Train attack model *A* that predicts membership(*x*,train set) from gradient norms and spectral signatures.
2. Extract gradient snapshots at unlearning boundaries (before/after)
3. Compute: $\| g_{before} \|_2 - \| g_{after} \|_2$ (gradient change magnitude)
4. Report: AUC of gradient-based MI attack (higher = unlearning failed)

Baseline Comparison: Output-logit MIA AUC from Varshney et al.

Our Proposal
Phase-1: Robust Verification

2. **Forensic Unlearning Membership Attacks (FUMA)**

Motivation: Unlearning leaves detectable behavioral artifacts even when logits match .

Method:

1. Create synthetic "canary" samples targeting feature k (e.g., extreme values)
2. Feed canaries to model pre/post-unlearning
3. Measure: $\Delta pred = | p_{before}(y | canary) - p_{after}(y | canary) |$
4. If $\Delta pred$ is systematically different for unlearned vs. never-trained features, FUMA succeeds.

Report: FUMA detection rate (should be <5% after unlearning vs. >50% without unlearning)

Our Proposal
Phase-1: Robust Verification

3. **Shadow Model Attacks**

Motivation: Attacker trains alternate models with/without target features .

Method:

1. Use historical checkpoints (active party stores per Varshney et al.) to retrain shadow models.
2. Shadow models are identical except trained on (retain set) vs. (retain set + target set).
3. Predict: Which dataset was used to train given model?
4. Report: Shadow MI AUC (attacks model, not membership)

Expectation: AUC $\approx$ 0.5 (unlearned model indistinguishable from retain-only model)

Our Proposal
Phase-2 : Feature Entanglement Analysis

- **Jacobian-Based Influence Matrix**

Motivation: Features are not independent; removing Xk may harm other parties' encoders .

Method: During training, at each epoch, compute low-rank Jacobian approximations:

- ρk,j(l)= // top singular value of (∂ hj(l)/ ∂ Xk) // 2 / // all singular values // F
- Store rank-*r* approximation: J≈UΣV*T* (reduced memory).
- For each (passive-party *j*, feature-set *k*) pair, compute:

  Influence*k*→*j* =median(*ρk*,*j* (*l*)) over all layers *l*

Classification:

- Low influence: *ρ*<0.15 → Unlearning should not degrade party *j*
- Medium influence: 0.15⩽ρ<0.40→ Monitor party *j* accuracy post-unlearning
- High influence: ρ⩾0.40 → Warn that party *j* may require local retraining

Our Proposal
Phase-3: Communication Efficiency

Experiments:

- Vary $d \in \{64,128,256,512\}$ (embedding dimensions)
- Vary $|D| \in \{1000,10000,100000\}$ (dataset size)
- Vary $uep \in \{5,10,20\}$ (unlearning epochs)
- Measure: Total communication bytes and wall-clock time

Report:

- Communication vs. d : Should scale roughly as dlogd
- Communication vs. $|D|$: Should scale linearly

Comparison to Full Retraining

- Baseline: Full retraining from scratch
- Speedup=Tretraining/Tunlearning

Our Proposal
Phase-4: Iterative Unlearning Robustness

Motivation: Real deployments involve repeated unlearning requests. Does utility/privacy degrade cumulatively?

Protocol:

1. Train VFL model on full dataset
2. Round 1: Unlearn top-5 features (selected by influence magnitude). Measure: accuracy, MI AUC
3. Round 2: Unlearn next 5 features. Measure: accuracy, MI AUC
4. Round 3: Unlearn next 5 features. Measure: accuracy, MI AUC
5. Round 4: Unlearn next 5 features (total 20 features). Measure: accuracy, MI AUC

Report: Does ϵtotal remain within acceptable bounds after 4 sequential unlearnings.

Literature and Dataset

Base Paper:
Unlearning Clients, Features and Samples in Vertical Federated Learning
PDF: **https://petsymposium.org/popets/2025/popets-2025-0048.pdf**

Others :

- Sidiropoulos, E., et al. (2025). "Unlearning-Specific Membership Inference Attacks." Proceedings of the Conferenceon Empirical Methods in Natural Language Processing (EMNLP).
- Sidiropoulos, E., et al. (2025). "Identifying Unlearned Data in LLMs via Membership Inference." Findings of the Association for Computational Linguistics: EMNLP 2025
- Thanh Tam Nguyen, Thanh Trung Huynh, Zhao Ren, Phi Le Nguyen, Alan Wee-Chung Liew, Hongzhi Yin, and Quoc Viet Hung Nguyen. 2025. A Survey of Machine Unlearning.
- Ziyao Liu, Yu Jiang, Jiyuan Shen, Minyi Peng, Kwok-Yan Lam, Xingliang Yuan, and Xiaoning Liu. 2024. A Survey on Federated Unlearning: Challenges, Methods, and Future Directions.

Datasets:

- MIMIC-III: **https://physionet.org/content/mimiciii/** (46K ICU patients)
- UCI Adult: **https://archive.ics.uci.edu/ml/datasets/adult** (32K census records)
- German Credit: Public UCI dataset (1K records)

THANK YOU